VARDHAN MISTRY

M.Sc. Mechatronics

+49 176 29028526
mistryvardhan@gmail.com
Chemnitz, Germany
vardhan-mistry
Vardhan1303
Portfolio Website

Profile Summary

M.Sc. Mechatronics Engineer with 2+ years of industrial experience in Computer Vision and AI/ML. Experienced in developing and deploying production-ready intelligent systems in automotive and industrial domains.

Professional Experience

04/2025 – 12/2025
Chemnitz, Germany

Master's Thesis
IAV GmbH
Focus: Camera Soiling Detection & Robustness Validation | **Grade: 1,0**

- Developed an end-to-end framework for camera soiling detection under adverse visual conditions using **PyTorch**.
- Designed **MudGAN** for synthetic dirt generation to support data-driven robustness training.
- Implemented and evaluated CNN (DeepLabv3+), Vision Transformer (SegFormer), and hybrid CNN (PIDNet) architectures for robust camera soiling detection.
- Built benchmarking and evaluation pipelines across pinhole and fisheye geometries.
- Optimized inference (7 ms) using **ONNX** on NVIDIA GPUs for real-time application.
- Presented production-ready prototype to **automotive and defense** stakeholders.

12/2024 – 02/2025
Chemnitz, Germany

Software Developer Intern
IAV GmbH
Focus: Synthetic Rain Augmentation

- Engineered a procedural raindrop generation tool for data augmentation in ADAS applications.
- Investigated synthetic degradation modeling to simulate camera blindness scenarios.
- Designed controlled experiments to assess detection robustness under rain conditions.
- Performed quantitative performance analysis to guide improvements.

08/2022 – 08/2023
Mundra, India

Assistant Manager
Adani Group
Focus: Automation & Robotics Systems

- Managed automation execution for a large-scale greenfield industrial project.
- Integrated and commissioned **Fanuc industrial robotic systems**.
- Coordinated multidisciplinary teams to deliver complex system-level milestones.
- Developed PowerBI dashboards for data-driven operational tracking and executive reporting.

Technical Skills

Programming: Python (NumPy, Pandas, PyTorch, TensorFlow, Keras), C++, MATLAB

AI & CV: CNNs, Vision Transformers (ViT), GANs, Object Detection, Segmentation, RAG, LLMs

MLOps & Cloud: Docker, Git, CI/CD, ONNX, TensorRT, CUDA, Model Quantization, Azure, REST API

Systems & HW: Linux, ROS/ROS2, NVIDIA GPUs (Jetson), STM32, Pinhole/Fisheye Camera Systems

Tools & Methods: Agile (Scrum), Jira, SAP MM, PowerBI, Project Management

Education

09/2023 – 01/2026
Weingarten, Germany

Master of Science in Mechatronics | Grade: 1,9
University of Applied Sciences Ravensburg-Weingarten (RWU)

07/2018 – 05/2022
Vadodara, India

Bachelor of Engineering in Mechanical | Grade: 1,2 (German Scale)
The Maharaja Sayajirao University of Baroda (MSU)

Languages

English: C1 – Fluent | **German:** B1 – Intermediate | **Hindi:** Native | **Gujarati:** Native